Criterion 7: Facilities and Technical Support (100)

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7.1. Adequate and Well-Equipped Laboratories, and Technical Manpower (50)

A) Laboratory Facilities

- **Number and Types of Laboratories**
  - **Core discipline-specific labs (e.g., Programming Lab, Electrical Lab, Mechanical Workshop, Data Science Lab, Electronics Lab).**
  - **Specialized labs for advanced courses (e.g., AI/ML Lab, IoT Lab, Cloud Computing Lab, VLSI Design Lab).**
  - **Research and project labs to encourage innovation and final-year projects.**
- **Adequacy & Alignment with Curriculum**
  - **Each lab mapped to relevant courses in curriculum.**
  - **Student–equipment ratio within acceptable limits (ensuring hands-on experience for all).**
  - **Availability of licensed software, domain-specific tools, and simulation environments.**
- **Modernization & Upgradation**
  - **Advanced instruments (e.g., CNC machines, microprocessor kits, GPU servers, robotics kits).**
  - **ICT-enabled labs (smart boards, AR/VR integration, cloud access).**
  - **Continuous investment in replacing outdated equipment.**
- **Laboratory Practices**
  - **Lab manuals, safety protocols, and SOPs in place.**
  - **Demonstration of experiments through virtual labs where possible.**
  - **Integration of labs with research and consultancy activities.**

B) Technical Manpower

- **Support Staff Strength**
  - **List of laboratory assistants/technicians/workshop instructors.**
  - **Department-wise ratio of labs to technical staff.**
  - **Adequacy in handling multiple labs and ensuring smooth functioning.**
- **Qualification & Skills**
  - **Diploma/Degree in relevant disciplines.**
  - **Specialized certifications (e.g., Cisco CCNA, AutoCAD, MATLAB, PLC training).**
  - **Skilled in equipment handling, troubleshooting, and lab safety measures.**
- **Roles & Responsibilities**
  - **Setup and calibration of instruments before practical sessions.**
  - **Support in student projects, research activities, and consultancy.**
  - **Maintenance of laboratory stock register and issue logs.**
  - **Assisting faculty during lab classes, workshops, and hackathons.**
- **Capacity Building**
  - **Regular training/upskilling through FDPs, workshops, and industry exposure.**

    Encouragement for pursuing higher studies or certifications

Expected Documentation / Evidence:

- List of laboratories with name, area (sq. m), major equipment/software, mapped course(s).
- Photos of labs and infrastructure.
- List of technical staff with designation, qualifications, years of experience, responsibilities.
- Maintenance records (logbooks, AMC documents, calibration certificates).
- Utilization schedules (lab timetables).

List of Laboratories

S. No.	Name of the Laboratory	No. of Students per Setup (Batch Size)	Name of Major Equipment	Weekly Utilization Status (Courses for which the Lab is used)
1	Programming Lab	20	40 High-end PCs with IDEs, Compilers, Projector	Data Structures, DBMS, OOPS, Python Programming
2	Electrical Machines Lab	15	DC Motors, Transformers, Measurement Kits	Electrical Machines I & II, Power Systems
3	Mechanical Workshop	25	CNC Machine, Lathe, Milling Machines	Workshop Practice, Manufacturing Processes
4	AI & Data Science Lab	20	GPU Server, MATLAB, TensorFlow, Python Libraries	AI, ML, Deep Learning, Data Analytics
5	Electronics Lab	20	CRO, Function Generator, Microcontroller Kits	Digital Electronics, Embedded Systems

Technical Manpower Support

S. No.	Name of Technical Staff	Designation	Qualification	Laboratory Assigned
1	Mr. A. Kumar	Lab Instructor	B.Tech (CSE)	Programming Lab
2	Ms. R. Sharma	Technician	Diploma in Electrical Eng.	Electrical Machines Lab
3	Mr. V. Patel	Workshop Instructor	ITI + Diploma (Mech.)	Mechanical Workshop
4	Ms. P. Nair	Technical Assistant	M.Sc. (AI & Data Science)	AI & Data Science Lab
5	Mr. S. Khan	Lab Technician	Diploma in Electronics	Electronics Lab

7.2. Additional Facilities Created for Improving the Quality of Learning Experience in Laboratories (20)

Categories of Additional Facilities

A) ICT & Digital Enhancements

- Installation of Smart Boards, AR/VR Kits, Virtual Lab Simulations.
- E-lab manuals with QR codes for video demonstrations of experiments.
- Remote-access software (e.g., MATLAB Online, Virtual Machines, Cloud-based tools).
- Laboratory Learning Management System (LMS) integration (Moodle/Google Classroom).

B) Research & Innovation Support

- Mini-project corners / Innovation cells inside labs.
- Dedicated project racks/workbenches with soldering stations, IoT kits, 3D printers.
- Establishment of Industry-supported Centers of Excellence (CoE) (e.g., AI CoE, Automation Lab).
- Prototype development facilities for students' projects/hackathons.

C) Student Learning Comfort & Accessibility

- Ergonomic seating, proper ventilation, air-conditioning for conducive learning.
- Provision of high-speed Wi-Fi in all labs.
- Assistive devices (screen readers, Braille kits, adjustable workbenches) for differently-abled students.
- Extended lab hours (beyond timetable) with faculty/technical staff support.

D) Advanced Equipment / Software Beyond Curriculum

- GPU workstations, licensed software (ANSYS, AutoCAD, TensorFlow).
- Industrial-grade kits (PLCs, Robotics Arms, CNC simulators).
- Virtual Reality labs for immersive learning in mechanical/architectural courses.
- Cloud-based remote experimentation platforms.

E) Safety & Quality Enhancements

- Advanced lab safety gear (fume hoods, fire extinguishers, emergency showers).
- Digital monitoring of lab equipment usage and energy consumption.
- Implementation of ISO/SALSA/OSHA-compliant safety protocols.

Additional Facilities in Laboratories

S. No.	Name of the Facility	Details / Description	Purpose for Creating Facility	Utilization (How Students Use It)	Relevance to POs / PSOs*
1	Virtual Lab Simulations	Remote-access portal integrated with curriculum experiments (NPTEL/SWAYAM Virtual Labs)	To allow students to repeat and practice experiments beyond class hours	Students use platform after lab hours to reinforce practical learning	PO5 (Modern Tool Usage), PSO1
2	IoT & Robotics Kits	Arduino, Raspberry Pi, Sensor Modules, 3D Printer	To promote project-based learning and innovation in automation	Final-year and hackathon projects actively use these kits	PO3 (Design/Development), PSO2
3	Cloud-based MATLAB/ANSYS	Licensed remote access software for simulation and modeling	To support simulation learning and research outside labs	Students access from home/hostel for assignments and research	PO5 (Modern Tools), PO4 (Investigation)
4	Extended Lab Hours Policy	Labs remain open till 8:00 PM with faculty supervision	To encourage research, mini-projects, and entrepreneurship activities	Final-year and PG students utilize extended timings for projects	PO9 (Individual & Team Work), PSO3
5	Assistive Technologies	Screen readers, Braille kits, adjustable benches	To ensure inclusivity and accessibility for differently-abled students	Students with special needs use these for lab participation	PO6 (Engineer & Society), PO8 (Ethics)
6	Industry-Supported CoE	Center of Excellence in AI & Automation with industrial collaboration	To expose students to real-world industrial tools and practices	Used for internships, training, and industry-driven student projects	PO12 (Lifelong Learning), PSO2

Documentation for NBA

S. No.	Additional Facility	Description	Purpose / Benefit	Evidence (Photo/Doc)
1	Virtual Lab Simulations	Online access to remote experiments	Helps students perform experiments outside lab hours	Screenshot / Access link
2	IoT & Robotics Kits	Arduino, Raspberry Pi, 3D Printer	Enables innovation, project-based learning	Photos of projects
3	Cloud-based MATLAB	Remote access for data analytics	Supports anytime-anywhere learning	License invoice / Access portal
4	Extended Lab Hours	Labs open 7 PM – 9 PM	Encourages project & research work	Circular/Notice
5	Assistive Tools	Screen reader software	Supports differently-abled learners	Accessibility audit report

7.3. Maintenance of Laboratories and Overall Ambiance (10)

Components to Cover in Documentation

A) Laboratory Maintenance

- **Preventive Maintenance Practices**
  - **Annual Maintenance Contracts (AMC) with vendors.**
  - **Periodic servicing and calibration of equipment.**
  - **Lab logbooks recording usage, faults, and servicing.**
- **Infrastructure Maintenance**
  - **Repair schedules for furniture, electrical, plumbing.**
  - **IT support for system upgrades and software renewals.**
  - **Inventory audits (spares, consumables, lab kits).**
- **Hygiene and Safety**
  - **Regular cleaning of labs, dust-free environments for electronics labs.**
  - **Safety audits, fire extinguishers, fume hoods, first-aid kits.**
  - **Signage for safety instructions and emergency exits.**

B) Overall Ambiance of Laboratories

- **Physical Environment**
  - **Well-lit, ventilated, air-conditioned (where necessary).**
  - **Ergonomic seating and adequate spacing between setups.**
  - **Properly designed workbenches with power supply points.**
- **Learning Ambiance**
  - **ICT-enabled smart boards/projectors in labs.**
  - **E-lab manuals, QR codes for experiment videos.**
  - **Wi-Fi access and digital learning resources.**
- **Inclusivity & Accessibility**
  - **Facilities for differently-abled (ramps, adjustable benches, screen readers).**
  - **Gender-sensitive spaces and secure environment.**
- **Student Feedback and Continuous Improvement**
  - **Feedback system for students to report lab-related issues.**
  - **IQAC/Departmental committee regularly reviews lab conditions.**
  - **Improvements documented (e.g., upgrading PCs, replacing outdated kits).**

Evidence Required for NBA

- Maintenance logbooks (equipment usage, servicing records).
- AMC copies (annual maintenance contracts for major equipment).
- Calibration certificates (for precision instruments).
- Photos showing:
  - Clean and organized labs.
  - Safety signboards, fire extinguishers, first-aid facilities.
  - ICT-enabled ambiance (projectors, smart boards, Wi-Fi).
- Student feedback reports on lab ambiance.

Suggested Annexure Format

S. No.	Lab Name	Maintenance Practice (Logbooks/AMC/Calibration)	Ambiance Features (Lighting, Seating, ICT, Safety)	Evidence (Photo/Doc Ref.)
1	Programming Lab	AMC for PCs, Antivirus updates, Weekly IT checks	Air-conditioned, Smart Board, Wi-Fi enabled	AMC_2025.pdf, Photo1.jpg
2	Electronics Lab	Calibration of CROs, AMC for Kits	Dust-free, Fire Extinguishers, Safety Posters	CalCert_EL2025.pdf, Photo2.jpg
3	Mechanical Workshop	Machine servicing logbooks, AMC with vendor	Spacious, Ventilated, Safety Helmets & Aprons	Logbook_MW2025.pdf, Photo3.jpg
4	AI & DS Lab	GPU server maintenance, Software renewal logs	Wi-Fi, ICT-enabled, Ergonomic Seating	Renewal_License.pdf, Photo4.jpg

7.4. Safety Measures in Laboratories (10)

Key Safety Measures to be Documented

A) General Laboratory Safety

- **Fire Safety:**
  - **Fire extinguishers (CO_2, Dry Powder) placed in every lab and checked annually.**
  - **Fire alarms and smoke detectors installed.**
  - **Clearly marked emergency exits.**
- **Electrical Safety:**
  - **ELCB (Earth Leakage Circuit Breakers), MCBs, proper earthing in electrical labs.**
  - **Insulated wiring, surge protectors for sensitive equipment.**
  - **Regular electrical audits and load testing.**
- **First Aid:**
  - **First-aid boxes stocked in every lab.**
  - **Quick access to medical staff/clinic on campus.**
  - **Contact details of emergency numbers displayed.**

B) Lab-Specific Safety Measures

- **Electronics / Electrical Labs:**
  - **Insulated mats near high-voltage equipment.**
  - **Warning signboards and “Switch Off after Use” labels.**
  - **Use of circuit breakers and fuses.**
- **Mechanical Workshops:**
  - **Mandatory use of safety goggles, gloves, helmets, aprons.**
  - **Guard shields on machines (lathe, milling, CNC).**
  - **Emergency stop buttons on all machines.**
  - **Proper ventilation and exhaust for welding/fume processes.**
- **Computer & AI Labs:**
  - **UPS and backup generators for uninterrupted power supply.**
  - **Antivirus and cybersecurity measures for system safety.**
  - **Ergonomic seating and posture awareness posters.**
- **Chemical / Physics Labs (if applicable):**
  - **Fume hoods, chemical storage cabinets, spill kits.**
  - **MSDS (Material Safety Data Sheets) displayed.**
  - **Proper waste segregation (hazardous vs. non-hazardous).**

Training & Awareness

- Induction programs for new students on lab safety protocols.
- Periodic safety drills (fire drills, evacuation drills).
- Posters/signage with Do’s & Don’ts in every lab.
- Training for technical staff in first aid & equipment handling.

Documentation for NBA Evaluation

- Safety Audit Reports (internal + external).
- Calibration certificates for safety devices (fire extinguishers, ELCBs).
- Photos of safety installations (extinguishers, signage, first-aid kits).
- Circulars/Notices about safety drills and training programs.
- Student feedback on lab safety (covered in ambiance feedback too).

Suggested Annexure Table

S. No.	Lab Name	Safety Measures in Place	Evidence (Photo/Doc Ref.)
1	Programming Lab	Fire extinguisher, First-aid box, UPS backup, ergonomic seating, safety posters	Photo1.jpg, SafetyAudit2025.pdf
2	Electronics Lab	ELCB, insulated mats, warning signboards, first-aid kit, annual fire safety drill	Photo2.jpg, CalibrationCert_EL2025.pdf
3	Mechanical Workshop	Helmets, gloves, aprons, machine guard shields, exhaust fans, emergency stop buttons	Photo3.jpg, WorkshopSafetyLog.pdf
4	AI & Data Science Lab	UPS, antivirus protection, ergonomic chairs, fire extinguisher, awareness posters	Photo4.jpg, ICTSafetyNotice.pdf
5	Chemistry Lab	Fume hoods, chemical storage cabinets, spill kits, eyewash station, MSDS displayed	Photo5.jpg, ChemSafetyAudit2025.pdf

7.5. Project Laboratory/Research Laboratory (10)

Components to be Documented

A) Project Laboratory

- **Dedicated lab space for final-year UG/PG projects.**
- **Equipped with basic infrastructure + advanced tools to support multidisciplinary projects.**
- **Facilities like:**
  - **High-end PCs/workstations.**
  - **CAD/CAM/CAE software (MATLAB, ANSYS, AutoCAD, TensorFlow).**
  - **Rapid prototyping tools (3D printers, PCB fabrication kits).**
  - **Development kits (Arduino, Raspberry Pi, FPGA boards).**
- **Support for student projects, hackathons, innovation contests.**

B) Research Laboratory

- **Designed for faculty research and PG/PhD work.**
- **Equipment beyond curriculum:**
  - **Advanced instruments (spectrum analyzers, robotics arms, AI servers).**
  - **High-performance computing (HPC) clusters or GPU servers.**
  - **Licensed domain-specific research software.**
- **Access to databases, digital libraries, simulation platforms.**
- **Used for sponsored projects (DST, AICTE, industry-funded).**

Student & Faculty Engagement

- **Students use these labs for:**
  - **Final year projects.**
  - **Mini-projects and capstone projects.**
  - **Startups and incubation activities.**
- **Faculty use for:**
  - **Funded research projects.**
  - **Industry consultancy.**
  - **Guiding PhD scholars.**

Documentation for NBA

Provide evidence through:

- **List of Project / Research / CoE Labs with area, facilities, and utilization.**
- **Student project details (titles, year, faculty guides).**
- **Research outputs: Publications, patents, prototypes.**
- **Industry/Research collaboration MoUs.**
- **Photos/videos of students working in the labs.**
- **Startup/incubation records (if applicable).**

Example Table

S. No.	Name of the Laboratory / Centre of Excellence
1	Final Year Project Laboratory
2	AI & Data Science Research Lab
3	IoT & Smart Systems Centre of Excellence
4	Advanced Manufacturing & Robotics CoE
5	Embedded Systems Research Lab

What Should Be Included in the List

Only dedicated labs that serve projects, research, or CoE activities. Examples:

- **Project Laboratory**
  - **Final Year Project Lab**
  - **Mini-project and Prototype Lab**
- **Research Laboratory**
  - **AI & Data Science Research Lab**
  - **Embedded Systems Research Lab**
  - **Robotics & Automation Lab**
- **Centre of Excellence (CoE)**
  - **IoT & Smart Systems CoE (with industry partnership)**
  - **Cloud Computing & Cybersecurity Innovation Centre**
  - **Advanced Manufacturing & 3D Printing CoE**